Improved Training Sample Efficiency and Inter-Device Generalizability in Optical Coherence Tomography Fluid Segmentation via Foundation Models

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Introduction

- Optical coherence tomography (OCT) is an imaging technique enabling detailed visualization of retinal structure and pathology such as accumulated fluid.
- Automated fluid segmentation could contribute to better disease management and treatment decisions.
- Challenges: 1) Grading OCT B-scans is a time consuming and labor intensive process. 2) Different OCT manufacturers have different noise characteristics (Fig 1) and supervised model struggles to generalize [1]. Hence, it requires collecting annotation device by device.
- Foundation models (FMs) may be able to ease the challenges. In particular, we investigated the following questions.

Q1. Sample efficiency. How much labeled data for training segmentation models can be saved by leveraging foundation models?
Q2. Inter-device generalizability. What is the extent of out-of-distribution generalizability of foundation models? Additionally, how does the performance of FMs compare to the performance of the in-domain standard approach?

Datasets

- OCT B-scans were annotated for intraretinal fluid (IRF), subretinal fluid (SRF), and pigment epithelium detachment (PED).
- D_S (Spectralis dataset): This is for fine-tuning models on segmentation and evaluate model performances. Images were acquired in nAMD and DME clinical trials. The dataset size of train, validation, and test was 2568, 458, and 371, respectively.
- D_C (Cirrus dataset): This is for inter-device generalizability evaluation. The test set is of size 99. The images were acquired in an nAMD clinical trial.

nAMD, neovascular age-related macular degeneration; DME, diabetic macular edema.

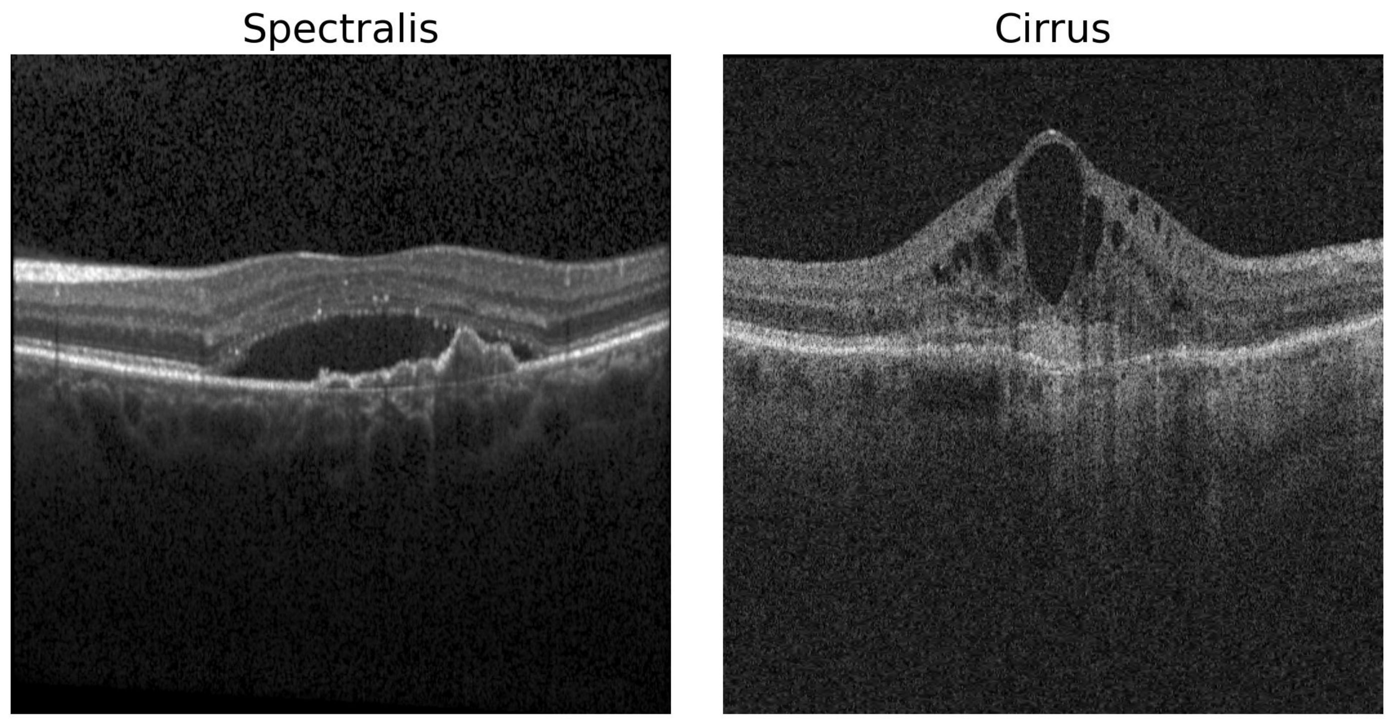
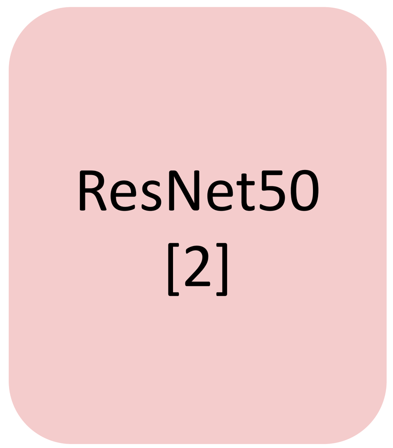
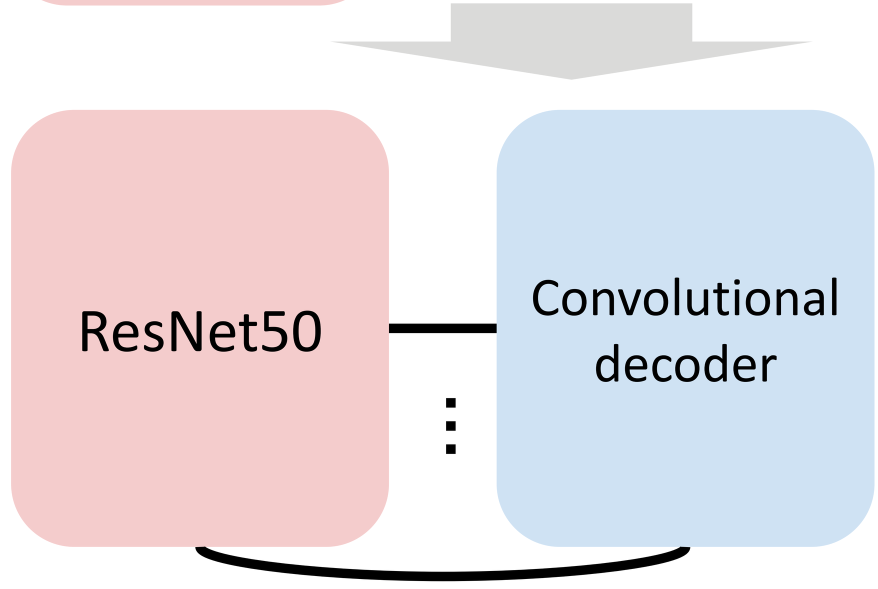


Fig 1. Sample image from Spectralis device and Cirrus device illustrating differences in noise characteristics.

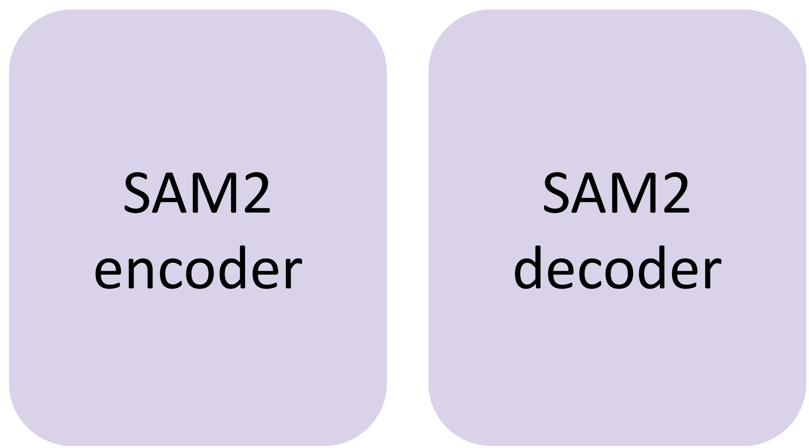
Methods



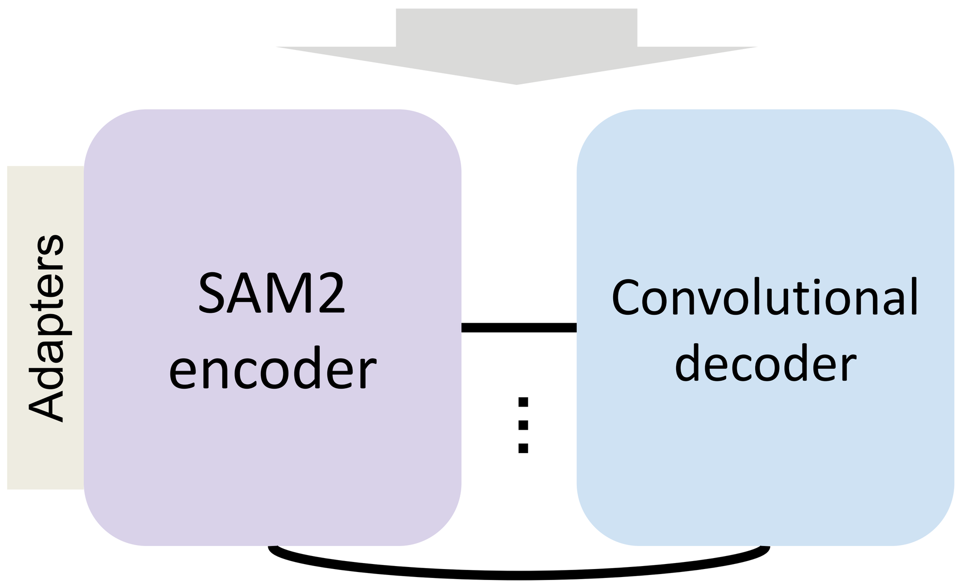
- Trained on internal OCT data comprised of over 18M B-scans following SimCLR method [3].
- Training image size of 224 by 224 pixels.



- SimCLR model was adapted to the segmentation problem by converting to ResUNet [4] with a randomly initialized decoder.
- Tips for fine-tuning: 1) Train on larger image size of 512 by 512 for small masks. 2) Assign a smaller learning rate to the encoder because decoder is randomly initialized.



- SAM2 is an FM for general-purpose segmentation developed by Meta Inc. [5].
- Trained on a large dataset of natural images and videos.



- The SAM2 encoder was connected to a convolutional decoder to form a SAM2-UNet architecture [6].
- Parameter-efficient fine-tuning was performed by incorporating adapters into the encoder while freezing the encoder.

Other experimental designs

- Hyper parameter tuning was performed using the loss on D_S validation set. 5 random seeds were run for each set of hyper parameters.
- Models with the best validation loss were evaluated without retraining on the test set of D_S and D_C.
- ImageNet pretrained ResNet 50 converted to ResUNet were fine-tuned on the training set of D_S and D_C separately as a comparator representing a standard approach. Fine-tuning protocol is the same as SimCLR ResUNet.

Results and Conclusions

A1. FMs require less data to achieve the peak performance.

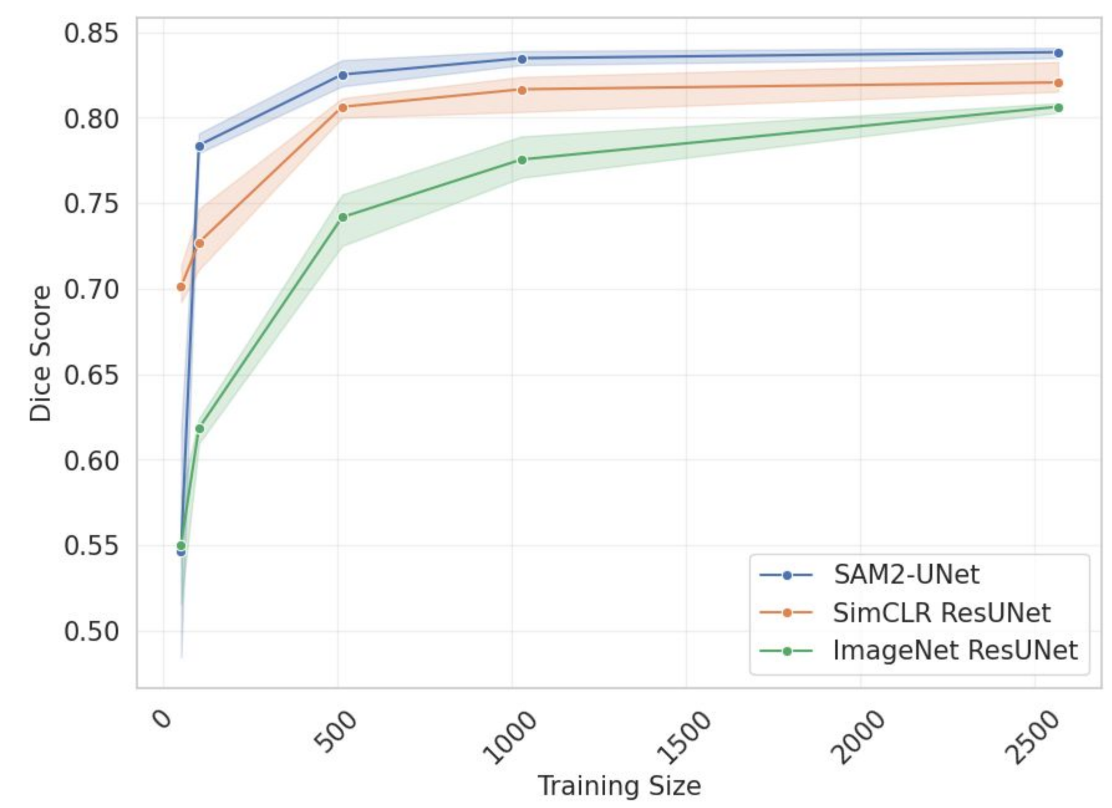


Fig 2. Average Dice score by training size. ImageNet ResUNet refers to the baseline model with pretrained ResNet50 on ImageNet. Dice score was average over 3 classes. ImageNet ResUNet starts declining at around 1000 while FMs maintain their performance level at around 500. SAM2-UNet outperforms SimCLR ResUNet except the very small data regime (~50).

A2. FMs are more robust to covariate shift due to differences in devices, exceeding in-domain standard supervised approach.

Model	IRF mean (min, max)	SRF mean (min, max)	PED mean (min, max)
SimCLR fine-tuned on D_S	0.71 (0.68, 0.75)	0.71 (0.67, 0.73)	0.64 (0.60, 0.70)
SAM2 fine-tuned on D_S	0.76 (0.73, 0.79)	0.72 (0.70, 0.74)	0.64 (0.60, 0.69)
ImageNet fine-tuned on D_S	0.69 (0.66, 0.71)	0.45 (0.39, 0.57)	0.42 (0.34, 0.49)
ImageNet D_C fine-tuned on	0.70 (0.65, 0.72)	0.55 (0.48, 0.67)	0.59 (0.57, 0.61)

Table 1. Inter-device generalizability results. Dice score on the test set of D_C by fluid type is shown. The statistics are over 5 seeds.

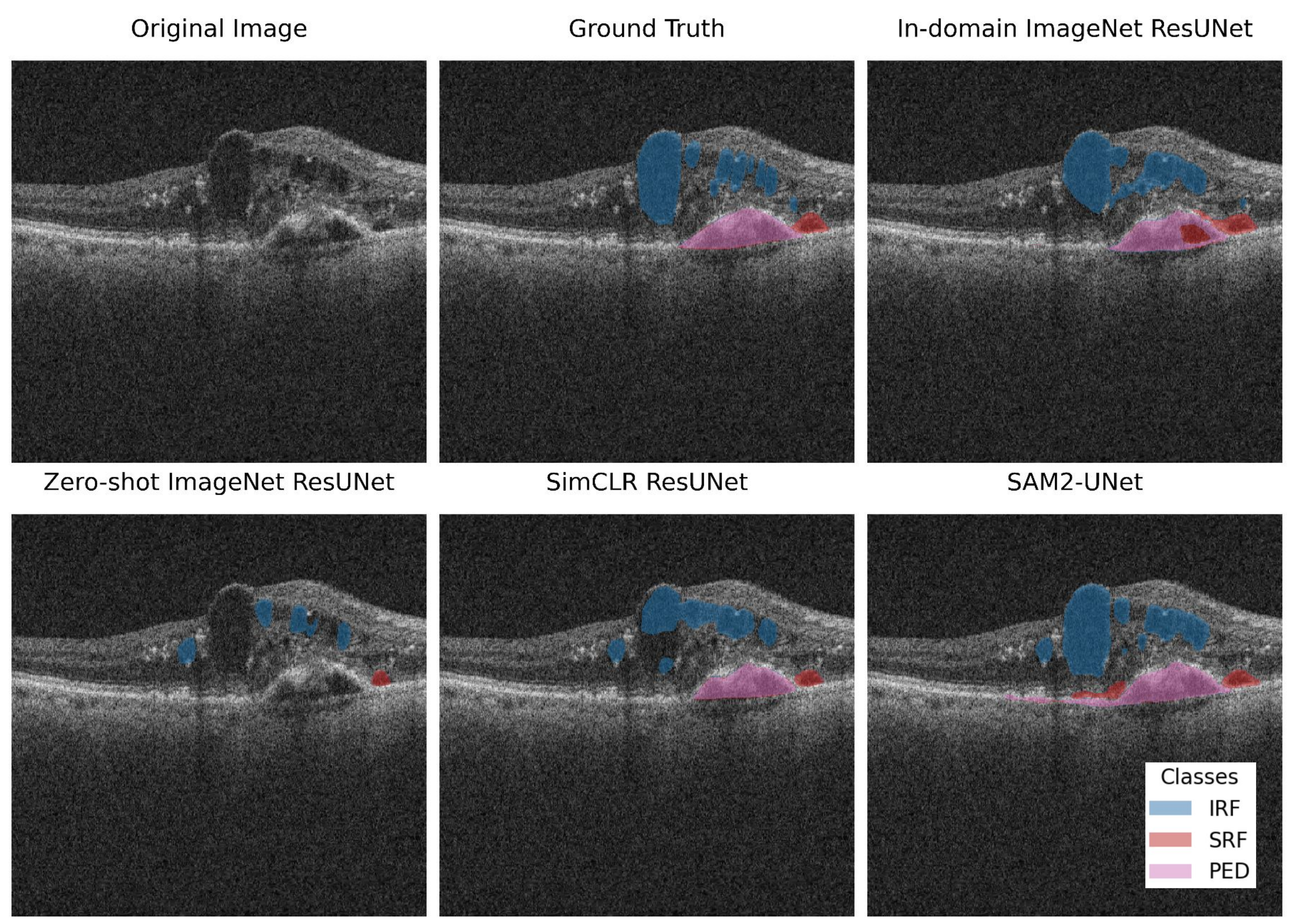


Fig 3. A representative case highlighting the qualitative differences between different approaches. Other models were trained on D_S and evaluated in Fig 2. The in-domain ImageNet ResUNet model violates topology of fluid and the zero-shot ImageNet ResUNet model misses significant portion of fluid.

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Study Disclosures

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